



ENDOC-LPI OBD-IIB

By Tecsage Innovation

Abstract

EnDoc stands as a versatile desktop solution for enhancing ST-LPI ECU management, offering reliability, platform independence, and ease of use.

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List of features in EnDoc – OBD IIB [Version 1.4]

1. Communication:

- ◆ Communication is carried through UART which can be connected through communication module in EnDoc.

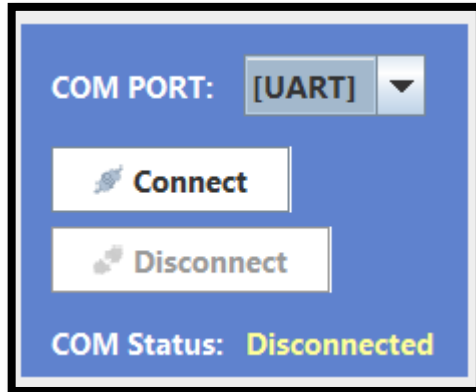


Figure 1: Communication module

- ◆ Communication module provides two major functionalities of connecting and disconnecting UART also it shows connection status.

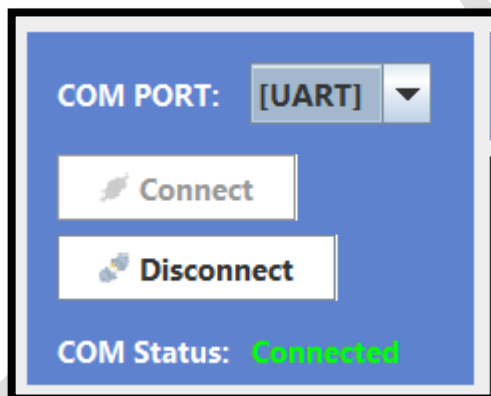


Figure 2: COM Connected

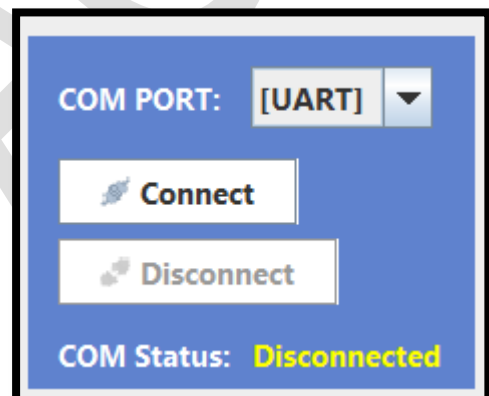


Figure 3: COM Disconnected

2. LUT Selection:

- ◆ This Module holds the functionality to choose embedded LUT based on different models of Engine and vehicles and flash selected LUT into the ECU

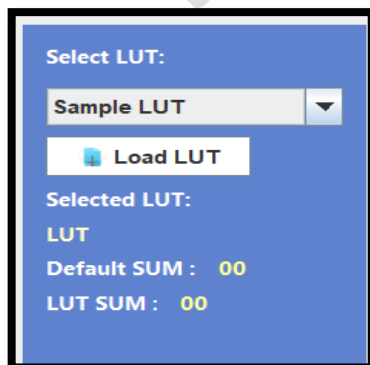


Figure 4: LUT Selection Module

- ◆ There are Eight Different LUT files which are embedded into the application which could be chosen from the drop-down menu

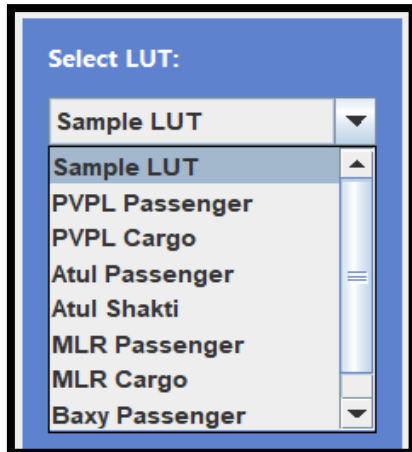


Figure 5: LUT Drop-down menu

- ◆ On clicking Load LUT button, selected LUT will be opened. Also, selected LUT, default SUM and LUT SUM will get updated dynamically
 - **Selected LUT:** It shows the currently selected LUT.
 - **Default SUM:** It shows the Actual SUM of LUT which is calibrated and tested.
 - **LUT SUM:** It shows the SUM of current LUT which is selected if it does not matched Default SUM it means LUT has been tampered.

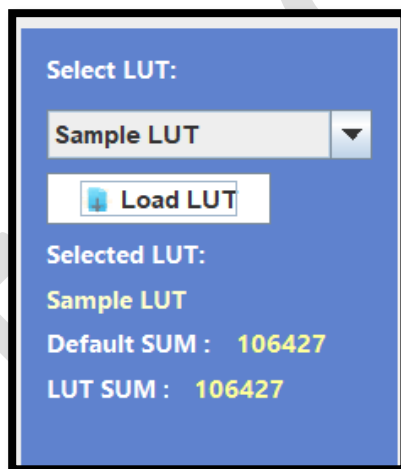


Figure 6: Selected LUT Information

3. LUT Reading from ECU:

- ◆ This module helps to read the flashed LUT from the ECU.

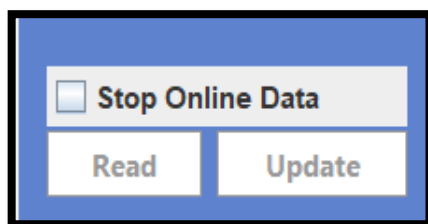


Figure 7: Read LUT Module

- ◆ Read LUT could be verified by the CVN which get updated dynamically when we read the LUT from the ECU.



Figure 8: CVN Module

4. LUT Module:

- ◆ This module shows all the selected LUT files and gives features as Updating LUT files, Saving LUT externally and open external LUT file
 - **Update LUT:** This module gives functionality to update values in selected LUT file
 - **Open LUT:** This module gives functionality to open LUT files that are saved externally
 - **Save LUT:** This module gives functionality to save LUT files externally

DOI

EGR

Compensation Factor

SOI

Title 1	Title 2	Title 3	Title 4	Title 5	Title 6	Title 7	Title 8	Title 9	Title 10	Title 11	Title 12	Title 13	Title 14	Title 15	Title 16	Title 17
Load/RPM	800	1300	1500	1800	2000	2200	2500	2600	2700	2800	2900	3000	3200	3500	4000	4500
15	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
20	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
25	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
30	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
35	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
40	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
45	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
50	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
55	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
60	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
65	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
70	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
75	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
80	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
85	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
90	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Update LUT

Open LUT

Save LUT

Figure 9: LUT Module

5. Flashing module:

- ◆ This module helps to flash the selected LUT file into the ECU. Also, it shows progress and process status while flashing LUT.

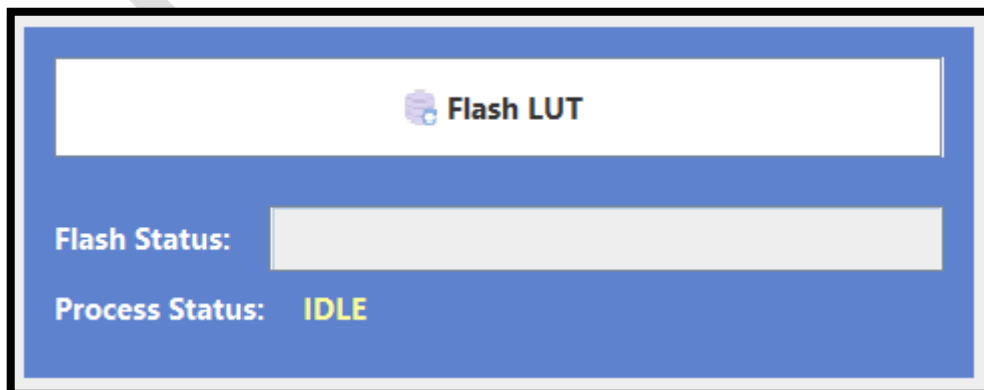


Figure 10: Flashing Module

- ◆ When Data is flashed Flash Status goes green in colour and process status reads “Data flashed”

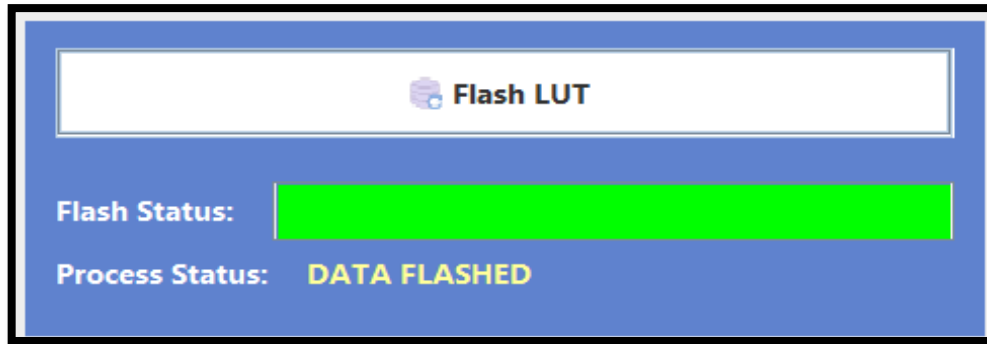


Figure 11: Data Flashed Module

6. Flashing Counter:

- ◆ This module helps to store the number of LUT flashed currently it records only session wise number of LUT flashing and total number of LUT flashed.
 - **Session wise recording:** It stores number of flashings done in current session i.e. till application started.
 - **Total Recording:** It stores number of flashings done till application was installed on the system.

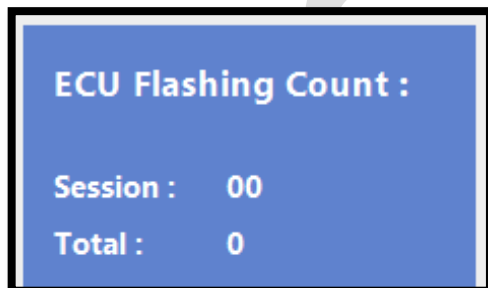


Figure 12: Flash counting module

7. MID Indicators:

- ◆ This module currently holds three main indications MIL, Pump and Engine status, which goes green when everything is working fine and goes red if respective DTC is found.
 - **MIL:** It stands for Malfunction Indication Lamp which is used to indicate if any malfunction is detected by the ECU.
 - **Pump Status:** This indication lamp is used to indicate working of Pump. If pump is working fine, it will glow green in colour and if any trouble is found related to pump it goes red.
 - **Engine Running Status:** This indication lamp is used to detect engine's running status, if engine is in running condition it will glow green and if Engine is in halt position it goes red.

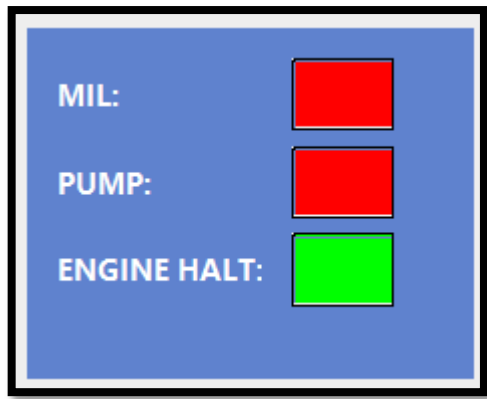


Figure 13: MID indications module

8. Level Display module:

- ◆ This module displays all the messages and level of water/solvent just like our display unit, also it shows current DTC count (if any) and functionality to Erase Live DTC codes.

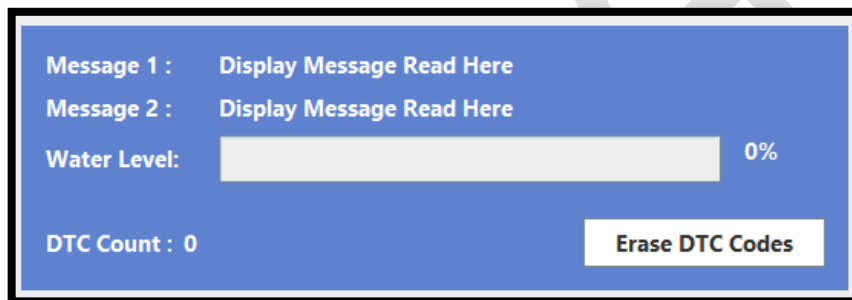


Figure 14: Level Display Module

9. Verification& VIN Flashing module:

- ◆ This module helps to verify the LUT using Calibration ID and CVN.
 - **Cal-ID:** It stands for Calibration Identity which is used to identify the LUT. It is of 16-Digit in length.
 - **CVN:** It stands for Calibration Verification number which is used to verify the LUT. It is automatically generated by the ECU.
- ◆ This module also has the functionality to flash VIN. VIN stands for Vehicle Identification Number which is used to verify the vehicle according to the LUT file flashed into its ECU. It is 17-digit in length.

Figure 15: LUT Verification and VIN Flashing Module

Note: Flash VIN button will get enabled automatically when 17-digit VIN is entered.
Barcode scanner could be used to scan VIN

10. Real-time Values:

- ◆ This module shows all the values from various sensors and actuators. It receives all the values from the ECU and could be used for diagnosis and calibration purposes. It shows data which are as follows:

- **Battery voltage**
- **TPS [%]**
- **ERPM**
- **EOT [°C]**
- **DOI [μS]**
- **Injections Per Second**
- **VS [Kmph]**
- **Solvent Pressure [kPa]**
- **EGR Set [%]**
- **Actual EGR [%]**
- **ΔP mBar**
- **EGT 1 [°C]**
- **EGT 2 [°C]**
- **Consumption Timer**
- **DTC Code**
- **EOT ADC**
- **EGR ADC**
- **Solvent Pressure ADC**
- **ΔP ADC**
- **EGT 1 ADC**
- **EGT 2 ADC**
- **TPS ADC**

Battery [V]	TPS [%]	ERPM	EOT [°C]	DOI [µs]	Inj per Sec	VS [km/h]	Sol. Pr [kPa]	EGR Set [%]	Actual EGR [%]	AP mBar	EOT 1 [°C]	EOT 2 [°C]	Cons. Timer	DTC Code	EOT ADC	EG
12.54	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2688	1
12.55	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2690	1
12.55	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2689	1
12.55	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2688	2
12.55	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2691	2
12.55	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2684	1
12.49	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2680	2
12.49	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2688	1
12.49	0	4248	25	1928	0	0	0	50	0	192	-39	-39	48		2693	1
12.49	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2689	1
12.49	0	4265	25	1928	0	0	0	50	0	192	-39	-39	48		2695	1
12.57	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2688	1
12.57	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2691	1
12.57	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2685	1
12.57	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2689	1
12.57	0	4248	25	1928	0	0	0	50	0	192	-39	-39	48		2694	1
12.51	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2688	1
12.51	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2683	1
12.51	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2690	1
12.51	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2691	1
12.51	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2688	1
12.52	0	4248	25	1928	0	0	0	50	0	192	-39	-39	48		2694	1
12.52	0	4265	26	1928	0	0	0	50	0	192	-39	-39	48		2688	2
12.52	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2689	1
12.52	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2690	1
12.52	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2680	2
12.54	0	4248	25	1928	0	0	0	50	0	192	-39	-39	48		2692	2
12.54	0	4248	26	1928	0	0	0	50	0	192	-39	-39	48		2687	1
12.54	0	4231	25	1928	0	0	0	50	0	192	-39	-39	48		2692	1

Figure 16: Online Values module

Note: Table can be cleared and saved using File menu.

11. Engine Parameter module:

- ◆ This module holds various parameters related to Engine and could be modified when needed.

Engine Speed (RPM) @ Cranking:	600
Fuel Cut-Off Cond. @ Engine speed (RPM):	5500
DOI (microsec) @ Engine Cranking mode:	5000
Delay for Injection Operation (microsec) :	20
Sec. EOT (°C) :	35
Sec. EOT Compensation Factor:	30
Pump ON time (sec) :	1
Glow Plug ON time (sec) :	15
VSS Correction Factor (CF * 10) :	7
Max Acceleration :	2000
EGR POT ADC Max. :	2700
EGR POT ADC Min. :	950
Consumption Error Max Limit :	18
Consumption Error Min Limit :	2
WI Reset :	5
Min Consumption Qty (Q/4) :	40
Inducement ON/OFF	1

Figure 17: Engine Parameter Module

12. OBD module:

- ◆ This module is used for Diagnosis purpose. OBD stands for On Board Diagnostics. In this module we display DTC codes and Error respective to that DTC also, freeze frame data is displayed on real-time basis.
- ◆ OBD is a valuable tool for vehicle owners and technicians, providing important information about the health and performance of a vehicle's systems. By monitoring key parameters and providing diagnostic information, OBD helps to ensure that vehicles are running efficiently, reliably, and in compliance with emissions regulations.

The screenshot shows the OBD Module interface with the following sections:

DTC Error Codes

MIL Status	Pending DTC Count	Stored DTC Count	MIL ON Time [min]

Freeze Frame Data

DTC code	RPM	EOT [°C]	TPS [%]	Reductant Pr. [kPa]	Actual EGR [%]

DTC Error Description

Stored DTC Code	DTC Description

ΔP mBar

NOTE: ΔPr. is for future reference.

Buttons: Erase/Update DTC Codes, Clear Table

Figure 18: OBD Module

13.LUT Verification module:

- ◆ This module helps to verify the individual LUT file if it is tampered or not. It calculates the sum of all the values of LUT and then displays the respective sum.

The screenshot shows the LUT Verification module interface with the following data:

SUM of LUT:

DOI LUT :	584276
EGR LUT :	11450
Temp. LUT :	29827
Pressure LUT :	177
SOI LUT :	00
EGT LUT :	34718
ΔP LUT:	23075
Engine Param :	16924
Cal ID :	965
Total SUM :	701412

Figure 19: LUT Verification Module

14. Role based Login module:

- ◆ **User:** This role is for the EOL testing user who only need to flash and test the ECU that are currently in production.
- ◆ **Customer:** This role is for the testing team who involve themselves in the field of development and test durability of ECUs in different conditions.
- ◆ **Admin:** This role is for Application development team who can modify and update the scope of application.

15. DTC module:

- ◆ This module displays all the DTC codes respective to their error description and DBounce time.
- ◆ DTC codes can be added, deleted and updated only by the authorised person using login credentials.

S.No	P_Code	Description	Dbounce Time [mSec]
1	P0689	Battery Voltage Low	4000
2	P0690	Battery Voltage High	4000
3	P0120	TPS Open Circuit	4000
4	P0123	TPS Sensor For Circuit High	4000
5	P0335	Crank Sensor Open Circuit	4000
6	P0197	ET Circuit Low	4000
7	P0195	ET Open Circuit	4000
8	P0196	EOT Sensor Out of Range	4000
9	P0199	EOT Sensor Performance Error	4000
10	P0406	EGR Circuit High	4000
11	P0405	EGR Circuit Low	4000
12	P0400	EGR Open Circuit	4000
13	P0404	EGR Out of Range	4000
14	P0401	EGR Stuck Flow Insufficient	4000
15	P0402	EGR Stuck Flow Excessive	4000
16	P204D	Reductant Pr Sensor Circuit High	4000
17	P204A	Reductant Pr Sensor Open Circuit	4000
18	P204B	Reductant Pr Sensor Out of Range Circuit / Performance	4000
19	P20E8	Reductant Pr Sensor Too Low	4000
20	P2048	Reductant Injector Circuit Low	4000
21	P2047	Reductant Injector Open Circuit	4000
22	P203F	Reductant Level Empty	4000
23	P20F5	Reductant Consumption Too High	4000
24	P20F6	Reductant Consumption Too Low	4000
25	P0545 / P2032	Exhaust Temperature Circuit Low 1/2	4000
26	P0544 / P2031	Exhaust Temperature Open Circuit 1/2	4000
27	P0543 / P2030	Exhaust Temperature Out of Range 1/2	4000

Figure 20: DTC Module

Note: Login form to manipulate DTC codes could be found under Edit menu.

16. Solvent Consumption module:

- ◆ This module displays all the information related to consumption of solvent. Currently it displays Actual and demand consumption, Injector pressure, Level Delta, Consumption Timer, current water percent and water percent at zero minute.

Online Values	Engine Parameters	LUT Module	OBD Module	LUT Verification	DTC Module	Sol. Cons. Module	Additons
Consumption Data							
Actual Cons.	Demand Cons.	Injector Pr.	Level Delta	Cons. Timer	Current water %	Water % @zero min	
70	454	114	1	51	0	0	

Figure 21: Solvent Consumption Module

17.Generate Error module:

- ◆ This module is used to generate real time errors directly through application without removing any sensor/actuator

☐ Sol. Level % :	0	▲	▼
☐ TPS :	0	▲	▼
☐ EGR :	0	▲	▼
☐ Red. Pr Sensor :	0	▲	▼
☐ Delta Pressure :	0	▲	▼
☐ EOT :	0	▲	▼
☐ EGT 1 :	0	▲	▼
☐ EGT 2 :	0	▲	▼
		☐ Intermittent	
		☐ Intermittent	
		☐ Intermittent	
Generate Error		Reset	

Figure 22: Generate Error Module

18.User/Admin Login:

- ◆ This module helps us to login and edit or update confidential data such as DBounce time, DTC codes and Error descriptions when logged in as user while we can update

License validity by using Admin login. These modules can be accessed using menu bar.

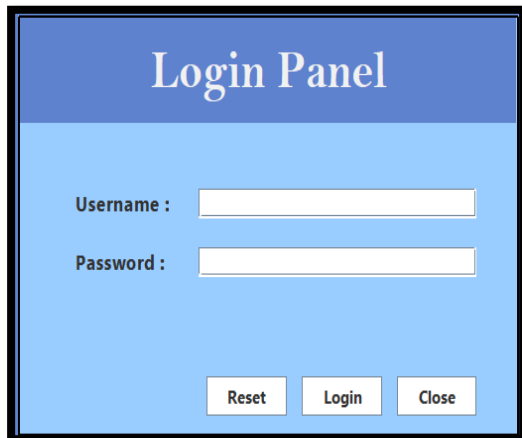


Figure 23: User Login Module

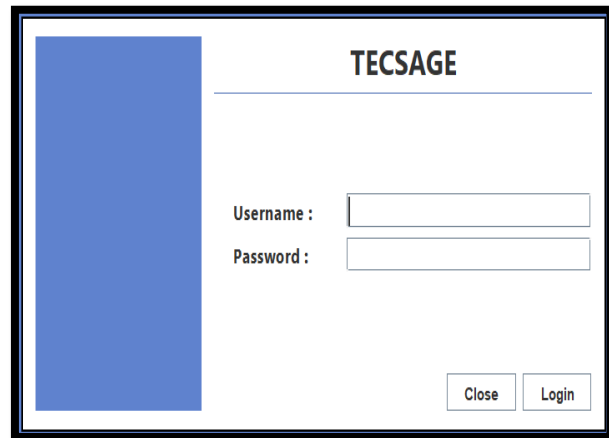


Figure 24: Tecsage Login Module

19.License:

- ◆ Licensing module shows the authenticity of application and all the details regarding application such as version, build and validity of license of application.

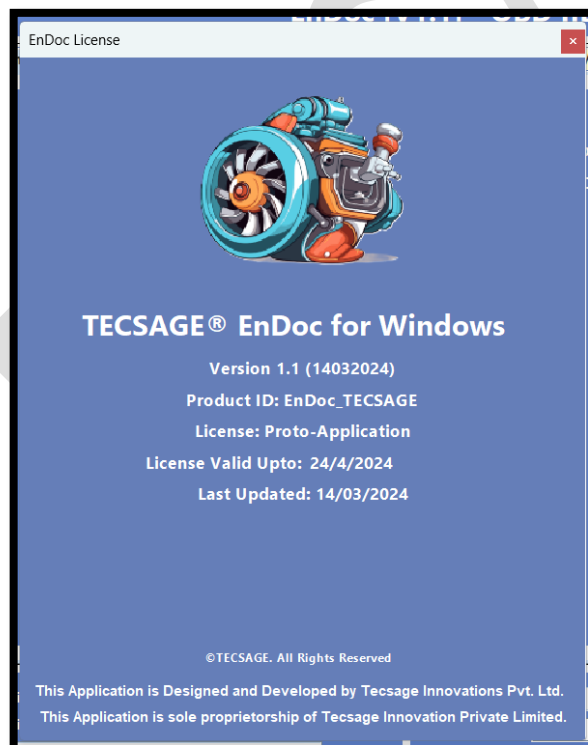


Figure 25: EnDoc License